Slide 1
CS 170
Java Programming 1
Eclipse@Home
Duration: 00:00:49
Advance mode: Auto

Notes:

Hi there. This is the CS 170, Java Programming 1 lecture that will show you how to download, install and customize the Eclipse Java development environment so that you can work at home. Because the version of Eclipse that we use for CS 170 has been customized, you'll need to follow these steps to make sure that your setup at home is the same as the one we're using in the OCC labs.

The version of Eclipse we're using this semester is version 3.4, also known as Ganymede. These instructions will work for Windows, Linux, Solaris and the Mac.

This lecture is also optional if you want to use the Computing Center for your assignments. Eclipse is already installed and configured there.

Slide 2
What is Eclipse?
Duration: 00:02:26
Advance mode: Auto

Notes:

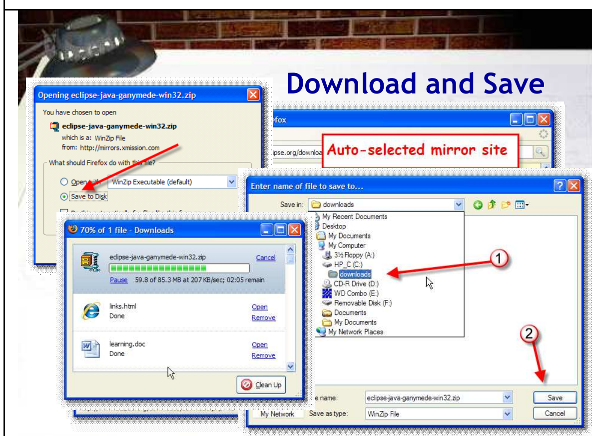
Eclipse is a full-featured Java development environment, designed for professional application development. Eclipse is really three tools in one: it's a Java development environment, a tool platform for creating other IDEs (such as C++), and an application framework for creating your own full featured GUI client applications, (called the RCP or Rich Client Platform).

Here are some of the main features of Eclipse:

- Like BlueJ, Eclipse is written in Java, so it works the same on Unix, on Mac OS X and on Windows. Eclipse uses its own "widget" toolkit, known as SWT (the Standard Widget Toolkit). Each version of the SWT uses native widgets so Eclipse looks pretty much like a native application.
- Eclipse integrates an editor, a compiler and a debugger, like BlueJ does, but it also has a host of features aimed at professional developers, such as support for team development and version control, and an array of "plugins" that can extend Eclipse to create mobile and Web applications.
- Like BlueJ and SciTE, Eclipse is free; it's also open-source, so you can modify it if you like, or write a plug-in to extend it to add some new functionality. (Both of those things go beyond what we'll be doing in this class, though.)
- Most of the advanced features in Eclipse aren't applicable to CS 170; what you will find interesting, though, is the incredible number of features Eclipse has to make writing your programs easier. In Eclipse:
- syntax errors are caught as you type, with a little red underline, much like spelling errors are pointed out in

Microsoft Word.

- you never have to compile your code: it's automatically compiled whenever you save.
- you don't have to remember what packages to import: the Eclipse Quick Fix feature will look them up for you.
- the Content Assistant will help you write your code, and the various wizards will walk you through creating a class, writing a constructor, or overriding a method.



Slide 3

Download and Save

Duration: 00:02:41

Advance mode: Auto

Notes:

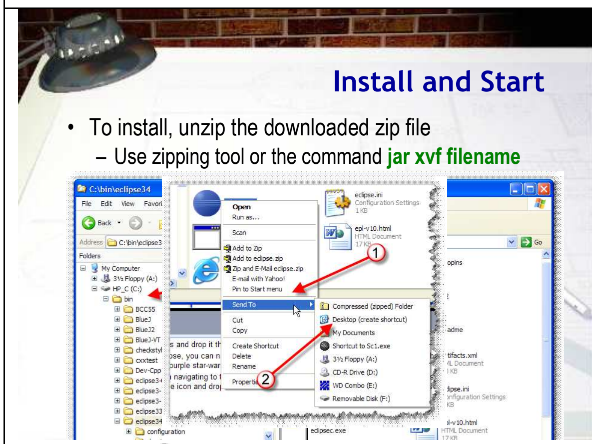
To download Eclipse, point your browser to http://www.eclipse.org which will take you to the Eclipse project home page that looks like this. Then, simple click the Download Eclipse button prominently displayed on the main page.

On the Eclipse Download page, first make sure you've selected the tab for Eclipse Packages (numbered 1 in the slide here). Then scroll down and find the section titled Eclipse IDE for Java Developers which I've labeled 2 here. You don't want the Java IDE for Java EE developers which is quite a bit larger and used for a different kind of Java development. Finally, in the area marked 3, click the link for the platform that you want to download. Since I'm taking these pictures on my PC, I'll click the link for Windows, but the steps are the same on the other platforms as well.

Once you've selected a package to download, you'll be taken to the "Mirror" page. Mirrors are trusted organizations such as universities, that donate server space and bandwidth so that the Eclipse Foundation main servers in Canada aren't overwhelmed with demand. Each time you select a download, one of the mirrors will be selected at random. If the selected site seems slow, you can select one of the other mirrors from the list below. (Watch out for the Google Ads that are inserted in the middle of the list, though.)

Downloading Eclipse. When the download starts, first click "Save to Disk" (not Run this File). Then, select a location on your hard disk where you want to save the file, making sure that you remember where you put it. I'm saving it in my C:\downloads folder, but you can save it anywhere. Also, make sure you remember the name of the file. Then, go out for a cup 'o' Joe. Eclipse is a very large program (about 85 MB) so it will take a little while to download. On broadband line, it should take about 10-15 minutes. If you find your download going really slow, you might want to switch to a different mirror site.

Unlike BlueJ and SciTE, Eclipse doesn't require the Java 2 SDK or JDK to run. All you need is a recent (version 5 or version 6) Java Runtime Environment or JRE. Eclipse uses its own compiler, not the javac compiler that's part of the JDK. Of course, if you've already installed the JDK, Eclipse is able to use that as well.



Slide 4

Install and Start

Duration: 00:02:30

Advance mode: Auto

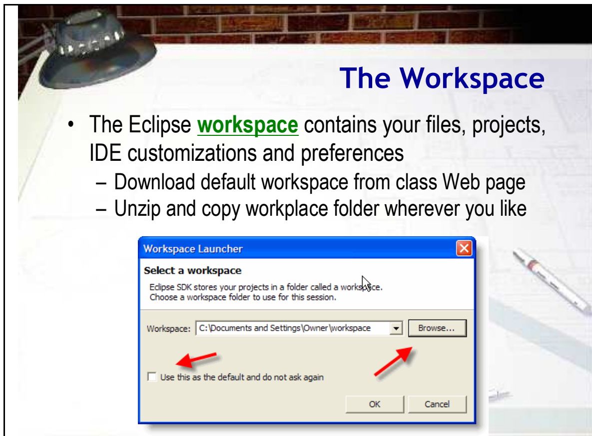
Notes:

"Installing" is kind of a misnomer, because unlike most other Windows programs, you simply, locate the zip file you downloaded and unzip it using a tool like WinZip or your can use the jar program that's included as part of the JDK. Simply open a command shell window, switch to the folder where you downloaded the file, and use the command jar xvf along with the filename. The name of the file that I downloaded for this lecture was eclipse-java-ganymede-win32.zip, but yours may be different. If you download the Mac version, you should be able to just double-click the tar-gz file to expand and unzip it..

Once you've unzipped the file, you'll find a folder called eclipse: Move that folder wherever you like in your file system. On my Windows computers, I put mine in a folder on the C:\ drive named bin (labeled in this screenshot as 1). If you have multiple versions of eclipse, you might want to unzip each version into its own folder. As you can see in the screenshot, I have different folders for versions 3, 3.1, and 3.1.1. For this version of Eclipse I've named the folder Eclipse34 (see the label 2). On my Mac, I just drag the whole Eclipse folder to Applications and drop it there. (You do need your admin password to do this, I think.)

To run Eclipse, you can navigate to this folder and just double-click the Eclipse icon labeled 3. It looks kind of like a little purple star-wars "death star", but it's supposed to be a solar eclipse.

Rather than navigating to the eclipse folder each time, you'll probably want to set up a shortcut. On the Mac, just drag the icon and drop it in the Dock. On Windows, right-click the icons and choose either 1) Pin to Start Menu if you want it to appear as an icon on the start menu, or 2) Send To Desktop if you want to create a shortcut on the desktop. Do not drag and drop the executable file on the desktop, though. It needs to stay inside its home folder.



Slide 5

The Workspace

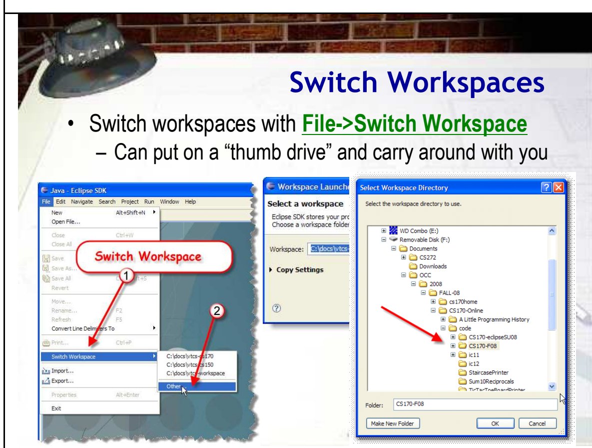
Duration: 00:01:28
Advance mode: Auto

Notes:

When you start Eclipse for the first time, you'll be asked to configure your own personal workspace. Your workspace will contain your files, as well as any customizations that you've made to Eclipse: If you click the checkbox on the left, Eclipse will only ask you to select a workspace this one time. If you leave it unchecked, you'll have to select your workspace each time you start the program.

At home you'll want to use the same customized workspace that we used in the labs. If you copied your OCC workspace file onto a USB thumb drive during class this week, you can just click the Browse button and select that now. If you want to have separate workspaces at school and at home, then download the default Virginia Tech workspace from the class Web page and unzip it onto your local machine. After you've unzipped it, you can move it, and, if you like, rename the workspace folder to something more memorable. Then, click the Browse button and select the newly downloaded workspace directory.

Once you've selected your workspace, Eclipse will start loading the various modules that make up the program. The first time you do this, it can take a couple minutes as your workspace is created. Subsequent startups will be much faster, however.



Slide 6

Switch Workspaces

Duration: 00:00:43
Advance mode: Auto

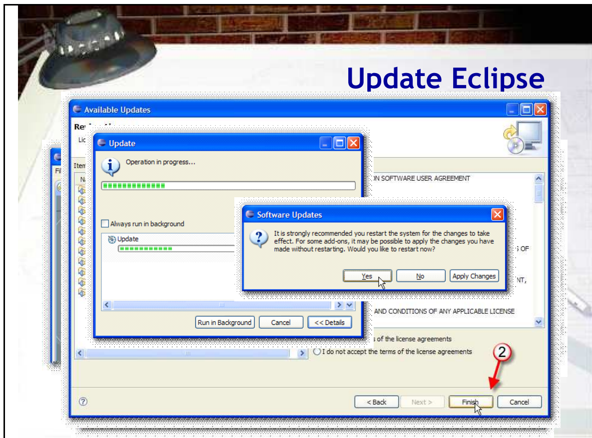
Notes:

At school, the computers are set to automatically load the default workspace off of the U: drive. If you want to share a workspace between your home computer and your school computer, you'll need to switch to the workspace on your USB thumb drive. Here's how.

Start Eclipse normally and when it starts, select Switch Workspace from the File menu. On the sub-menu, choose other.

When the Workspace Launcher dialog appears, just click Browse.

Then, select the folder containing your Eclipse workspace on your USB thumb drive like this.



Slide 7

Update Eclipse

Duration: 00:01:08

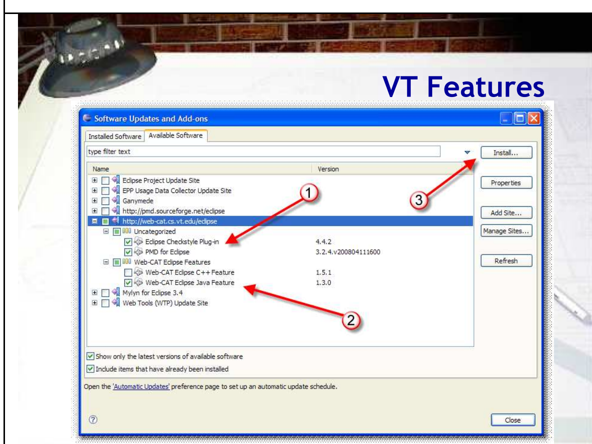
Advance mode: Auto

Notes:

The final step to complete your installation is to install the Virginia Tech customizations and to update Eclipse itself. You'll do both of these while Eclipse is running.

- First, choose Software Updates from the Help menu.
- On the Installed Software tab click the Update button to apply any patches developed since your version was released. Your computer will contact the main Eclipse servers and retrieve a list of newer modules that are available.
- When the list of available updates appear, you'll normally just accept them all and click the Next button.
- Finally, click the radio button marked 1 in the slide to accept the license terms. (All of Eclipse tools are released under an open-source license). Then, click Finish.

The updater should trudge along for awhile, downloading and installing the necessary updates. When it's done, you'll be asked to restart your system, but that just means restarting Eclipse, not rebooting your computer.



Slide 8

VT Features

Duration: 00:01:21

Advance mode: Auto

Notes:

To install the Virgina Tech customizations, such as the Web-Cat submitter and the Dr. Java, Checkstyle and PMD plugins, choose Software Updates from the Help menu once again. This time, though, you'll select the second tab, Available Software (instead of Installed Software) and then choose Add Site.

On the Add Site dialog, enter http://web-cat.cs.vt.edu/eclipse as the name of the site, and click OK.

Eclipse will then contact the Web-Cat site and download a list of updates. When the updates arrive expand the tree and select Checkstyle and PMD in the Uncategorized section (labeled 1 in the picture) and the Web-Cat Eclipse Java feature in the Web-Cat section. Once you've checked those, click the Install button. Accept the list of available updates, agree to the license and restart Eclipse again, just like you did when updating the other installed features.

Well, that's it. Now you can use Eclipse at home just like you do in the labs at school.